Batch-04 TECHNICAL PRESENTATION

Comparative Technical Analysis of Image Processing and Computer Vision

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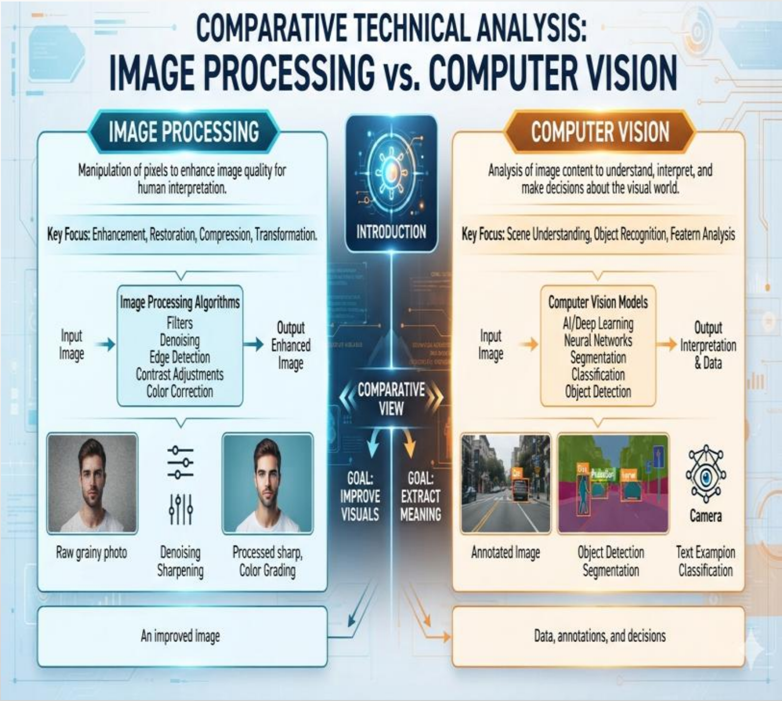
SIMATS ENGINEERING

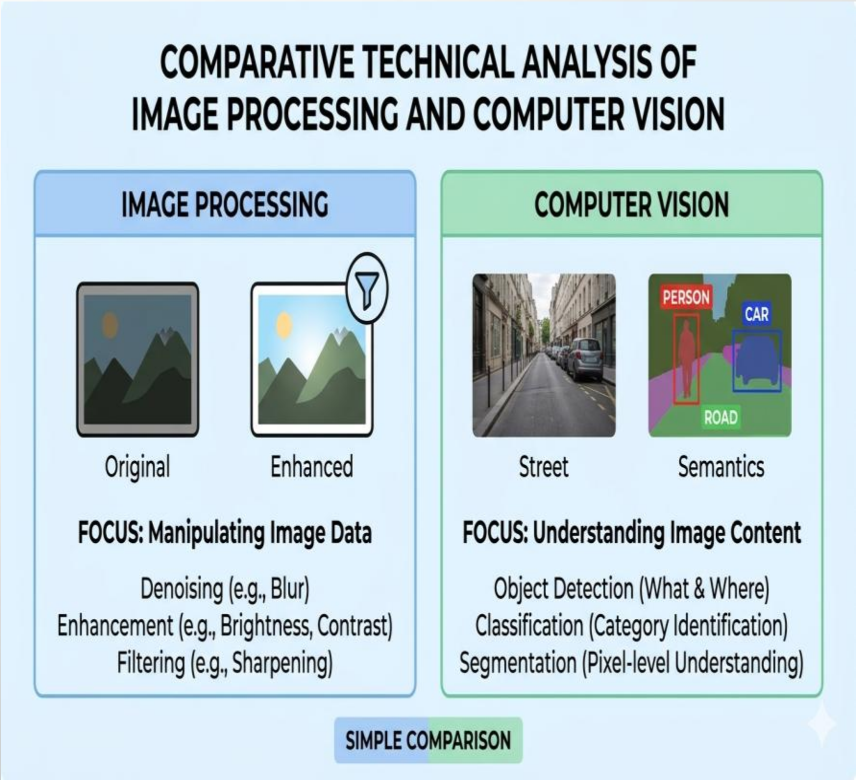
Introduction

- Focuses on improving or modifying digital images.
- Performs operations such as filtering, enhancement, restoration, and compression.
- Produces an improved version of the original image.
- Enables computers to understand the interpret images or videos.
- Uses Artificial Intelligence and Deep Learning to recognize objects and make decisions.
- Produces meaningful information rather than just an enhanced image.



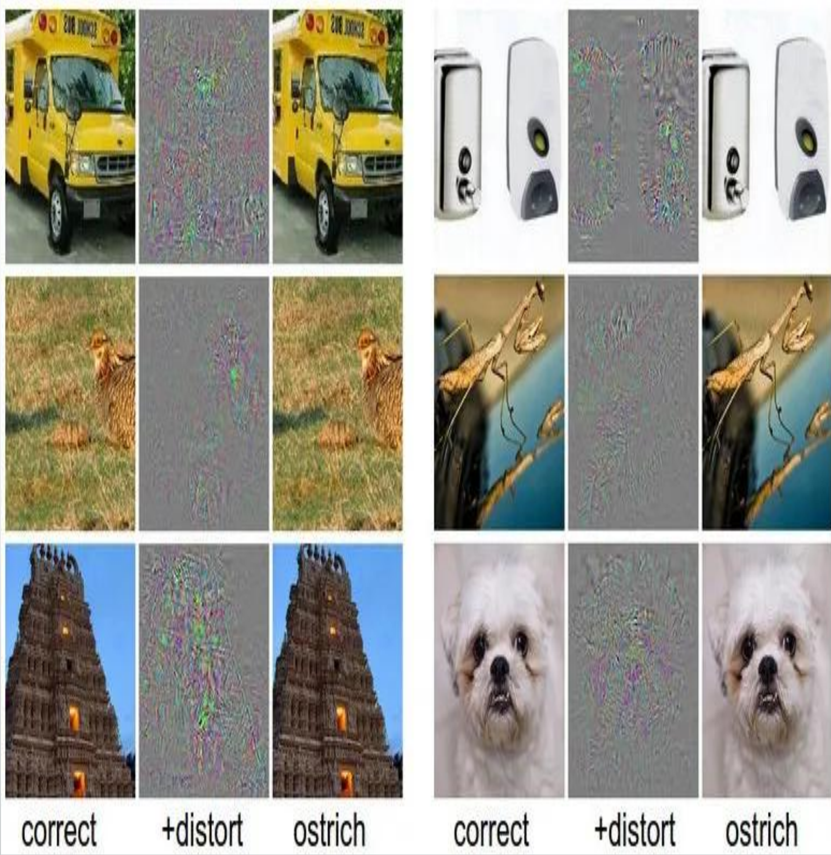
Objectives

- To understand the concept of Image Processing.
- To compare both technologies based on their scope and functionality.
- To analyze techniques and algorithms used in each domain.
- To study real-world applications.
- To identify advantages and limitations.
- To understand how Image Processing supports Computer Vision.

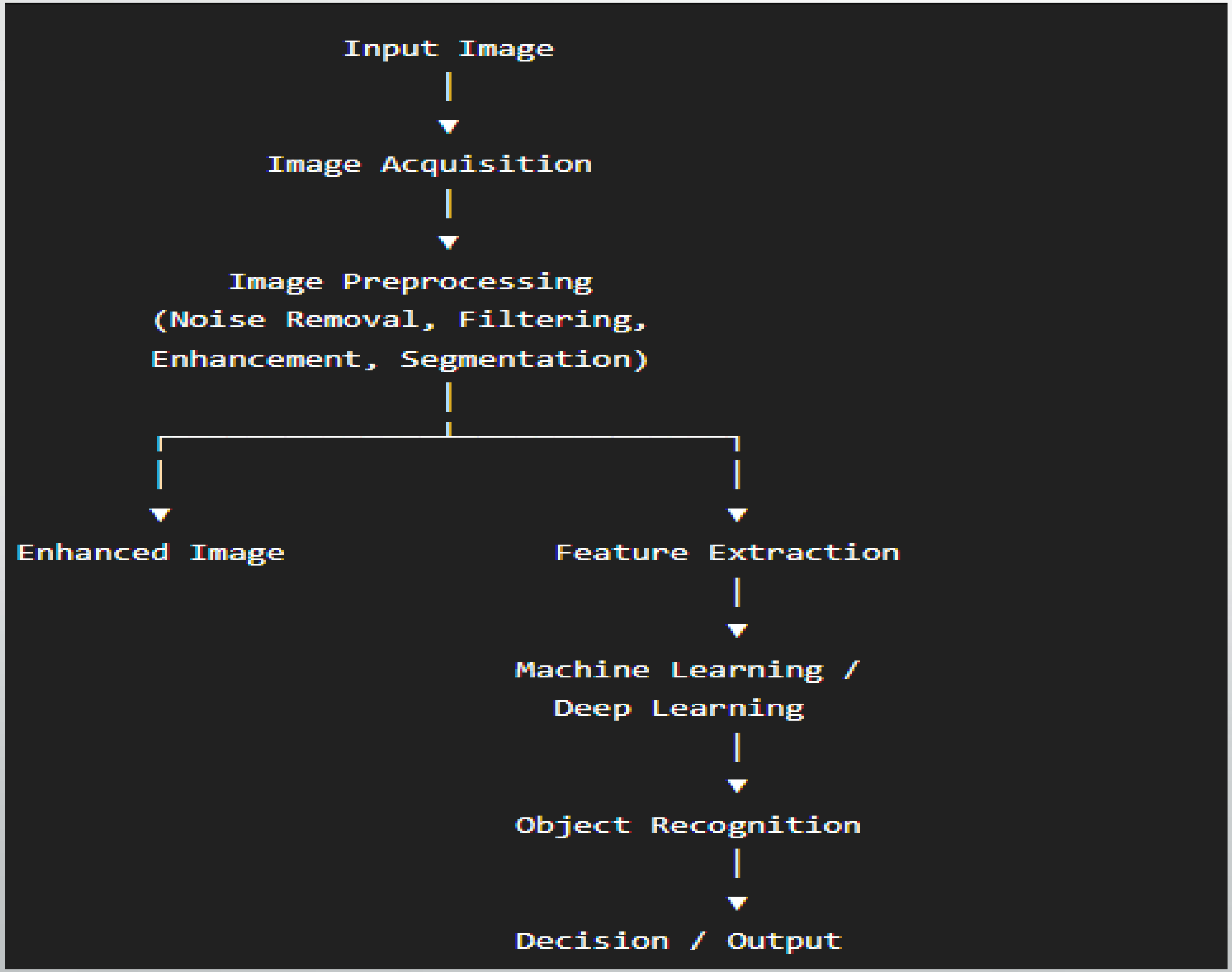


Problem Statement

- Modern applications such as healthcare, autonomous vehicles, surveillance, robotics, and industrial automation require computers to process and interpret visual information accurately.
- However, many people confuse Image Processing with Computer Vision, even though they solve different problems.
- This study aims to compare both domains by analyzing their objectives, workflows, techniques, outputs, applications, advantages, and limitations to clearly understand their roles in intelligent visual systems.



Architecture Diagram



Techniques & Algorithms

- **Image Processing Techniques**
- Image Filtering
- Histogram Equalization
- Edge Detection
- Thresholding
- Segmentation
- Morphological Operations
- **Algorithms**
- Gaussian Filter
- Median Filter
- Sobel Operator
- Canny Edge Detector

Computer Vision Techniques
Image Classification
Object Detection
Object Tracking
Face Recognition
Semantic Segmentation
Algorithms
CNN
YOLO
Faster R-CNN
SSD
ResNet
Vision Transformer (ViT)

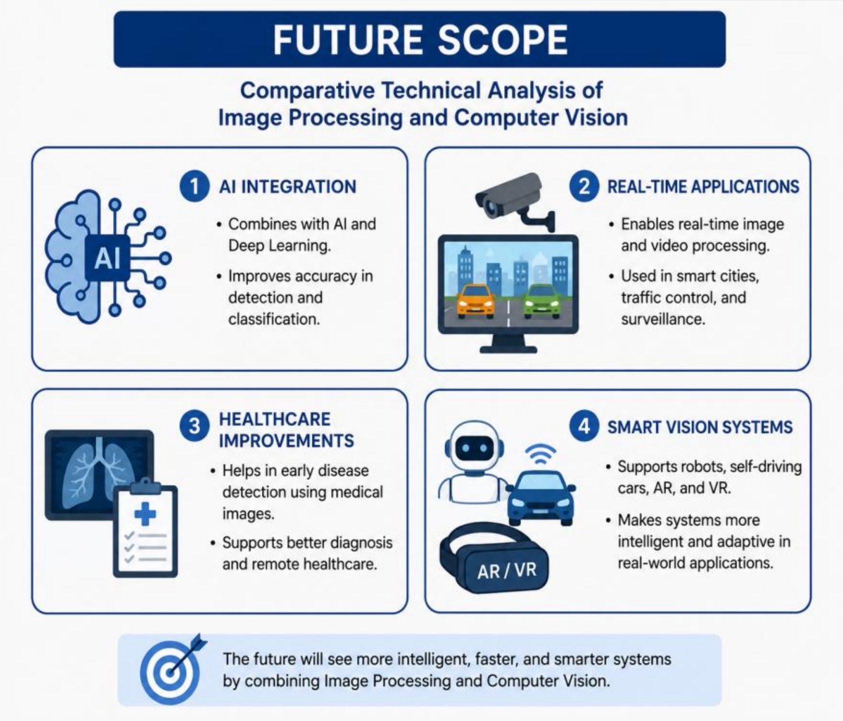
Applications

- **Applications of Image Processing**
- Medical Image Enhancement
- Satellite Image Processing
- Photo Editing
- Image Compression
- Fingerprint Recognition
- Document Scanning

- **Applications of Computer Vision**
- Self-driving Cars
- Face Recognition Systems
- Smart Surveillance
- Robotics
- OCR (Text Recognition)
- Medical Diagnosis
- Industrial Automation

FUTURE SCOPE

- **AI Integration**
  - Improve accuracy using AI and Deep Learning.
- **Real-Time Applications**
  - Used in smart cities, traffic control, and surveillance.
- **Healthcare Improvements**
  - Better disease detection using medical images.
- **Smart Vision Systems**
  - Support robots, self-driving cars, AR, and VR.



Conclusion

- Image Processing enhances and transforms digital images using filtering, segmentation, restoration, and enhancement techniques.
- Computer Vision uses Artificial Intelligence and Deep Learning to analyze images, recognize objects, and make intelligent decisions.
- Both technologies complement each other in modern applications such as healthcare, autonomous vehicles, security, robotics, and industrial automation.
- Understanding the differences between these domains helps in selecting the appropriate technology for solving real-world visual computing problems.


THANK
YOU